Proton NMR Spectroscopy Video Series 1

Video 3: Chemical Shifts

Video Series Goals

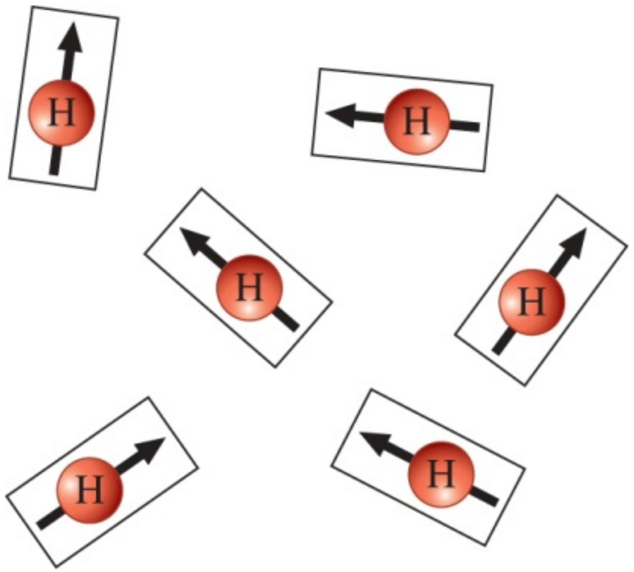
Video 1
- Introduction and Theory ☑

Video 2
- Anatomy of ^{1}H NMR spectra ☑
- Counting Unique Chemical Environments
- Signal Integrations

Video 3 **(Current)**
- Chemical Shift

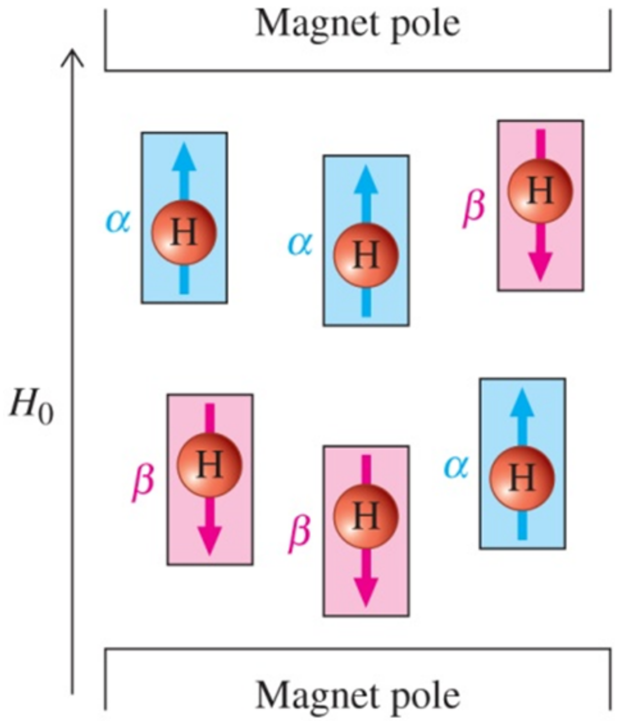
Video 4
- Splitting Patterns
- Coupling Constants

Review of NMR Theory



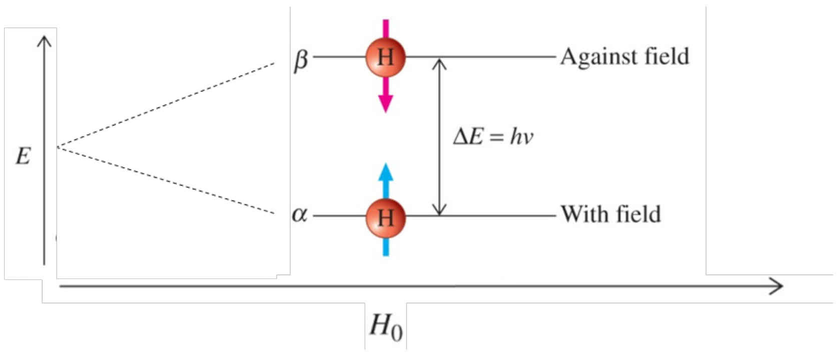
A

No external magnetic field: random orientation



B

In an external magnetic field: alignment with (α) or against (β) H_0

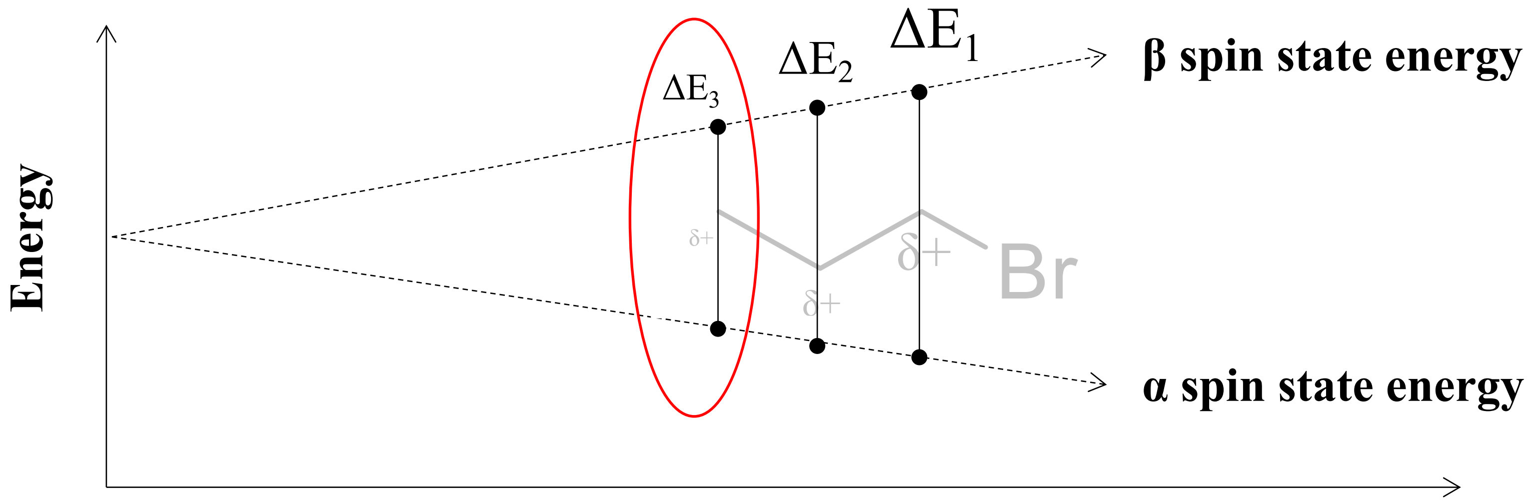


C

Energy difference between α and β spin states: depends on magnetic field strength and proton chemical environment

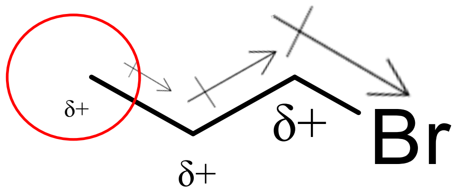
Figure adapted from Vollhardt, P.; Schore, N. Organic Chemistry Structure and Function. 7th ed.; W.H. Freeman: New York, 2014; p 381.

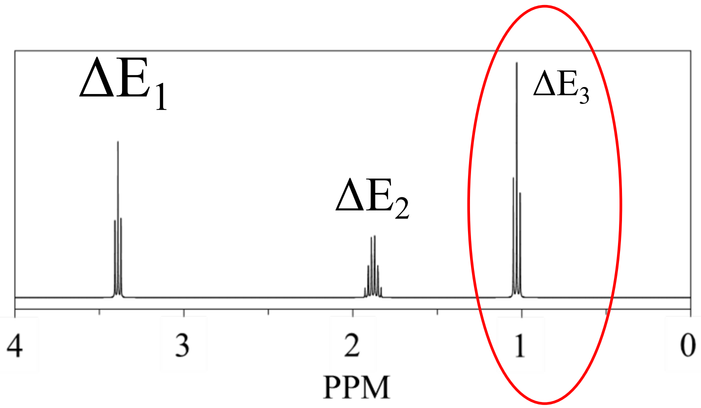
Review of NMR Theory



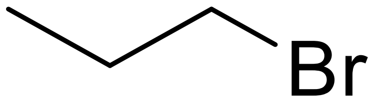
Magnetic field strength "felt" by protons
Depends on chemical environment

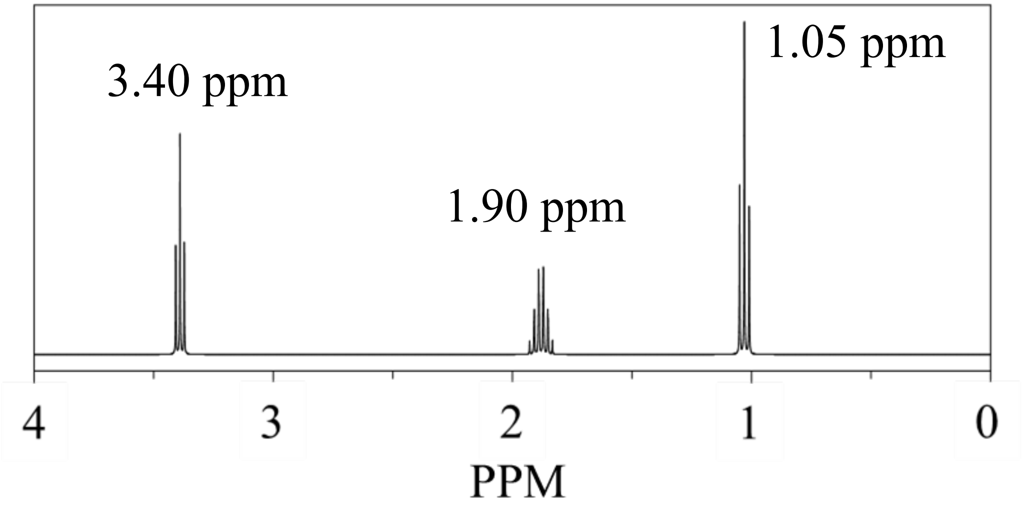
- Electron density **"shields"** protons
- Positively charged protons are **"deshielded"**





What is Chemical Shift?



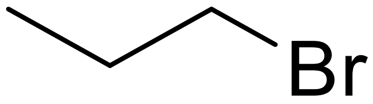


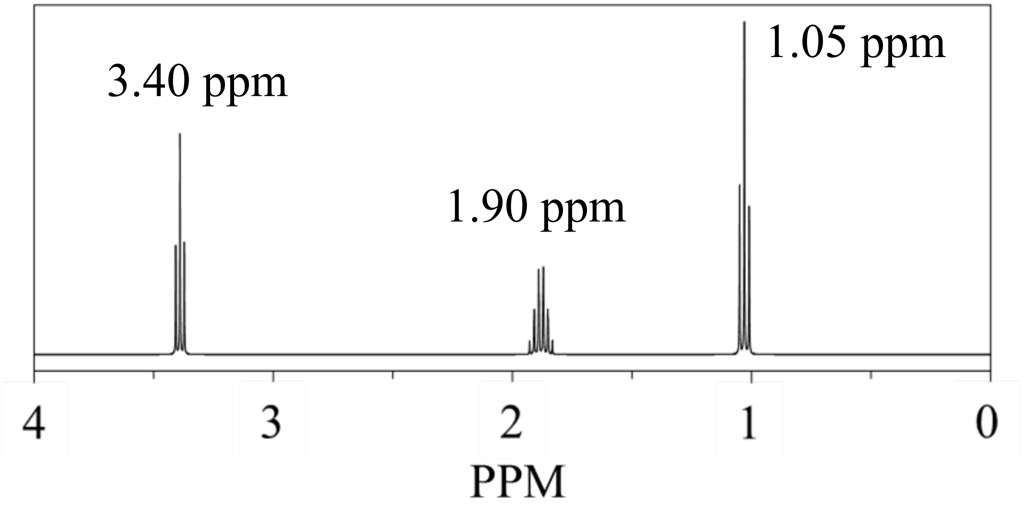
- Position of NMR signal along the x-axis
- Has units of δ (ppm)

$\Delta E = h\nu$

$$\delta\ (ppm) = \frac{Frequency\ of\ NMR\ signal\ (Hz) * 10^6}{Frequency\ rating\ of\ NMR\ spectrometer\ (MHz)}$$

What is Chemical Shift?





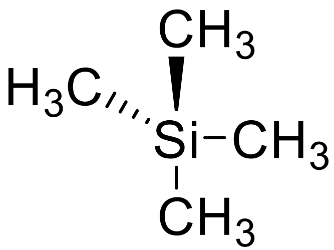
Increasing chemical shift

- Position of NMR signal along the x-axis
- Has units of δ (ppm)

$\Delta E = h\nu$

$$\delta\ (ppm) = \frac{Frequency\ of\ NMR\ signal\ (Hz) * 10^6}{Frequency\ rating\ of\ NMR\ spectrometer\ (MHz)}$$

- δ **increases from <u>right to left</u>**
- Scale is relative to tetramethylsilane



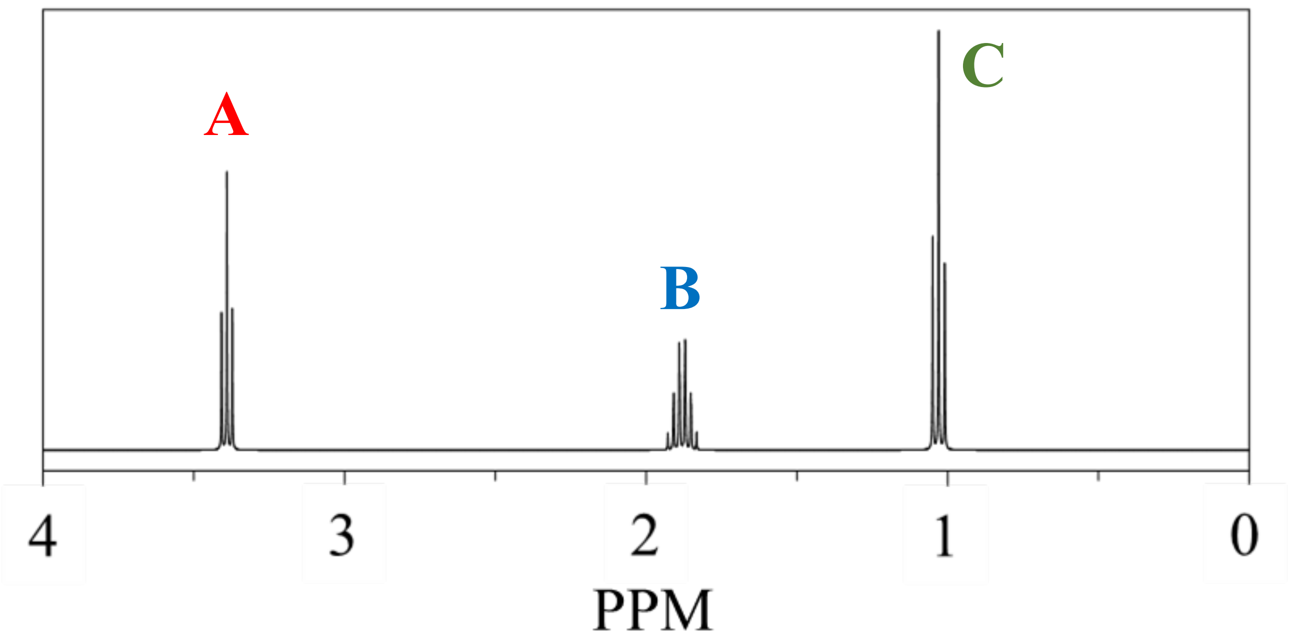
Has one ^{1}H NMR signal **defined as 0 ppm**

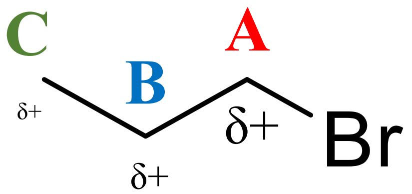
Chemical shift depends on a proton's chemical environment

Chemical Shift Terminology

"Downfield"
"Deshielded"
(less nearby electron density)

"Upfield"
"Shielded"
(more nearby electron density)





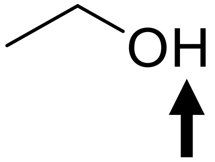
Factors that Effect Chemical Shift

Electron Density

- Electron donating and withdrawing functional groups in a molecule

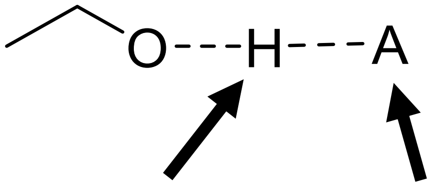
Hydrogen Bonding

No hydrogen bonding



Protic hydrogen

Hydrogen bonding



Protic hydrogen
“H-bond donor”

“H-bond acceptor”

R-O**H**	Hydroxyl	0.5 – 5 (often broad)
R-N**H**	Amino	0.5 – 5 (often broad)

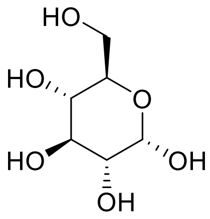
Factors that Effect Chemical Shift

Electron Density

- Electron donating and withdrawing functional groups in a molecule

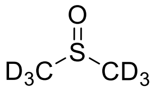
Hydrogen Bonding

Solvent Effects

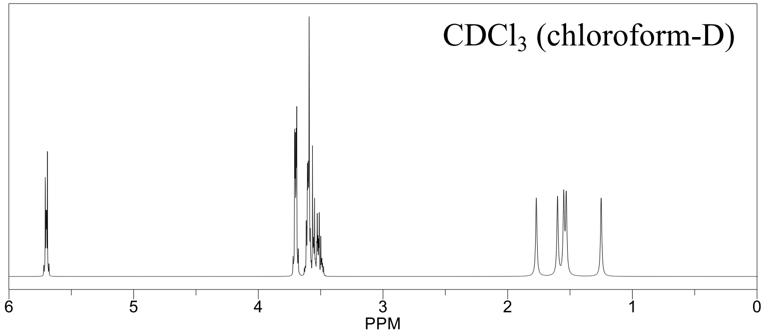


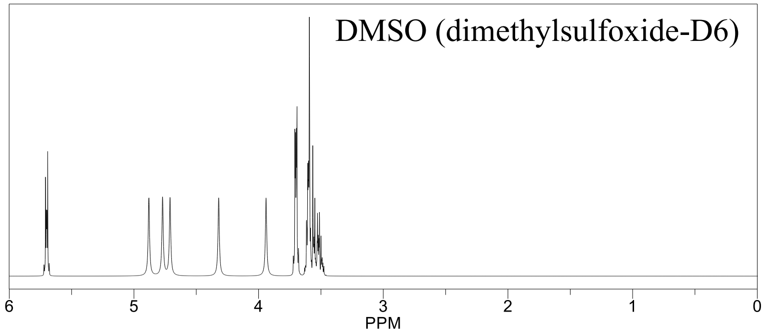
Chemical shifts in NMR spectra depend on the solvent used

- Solvent molecules interact with the sample!



Simulated ^{1}H NMR spectra of glucose





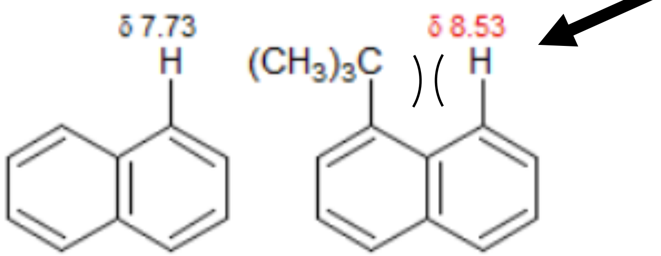
Factors that Effect Chemical Shift

Electron Density
- Electron donating and withdrawing functional groups in a molecule

Hydrogen Bonding

Solvent Effects

Steric Compression



Steric compression of this proton causes electron cloud deformation

Figure (accessed June 8, 2020) https://www2.chem.wisc.edu/areas/reich/nmr/05-hmr-02-delta.htm

Factors that Effect Chemical Shift

Electron Density

- Electron donating and withdrawing functional groups in a molecule

Hydrogen Bonding

Solvent Effects

Steric Compression

Magnetic Anisotropy

Hybridization (sp^3, sp^2, sp)

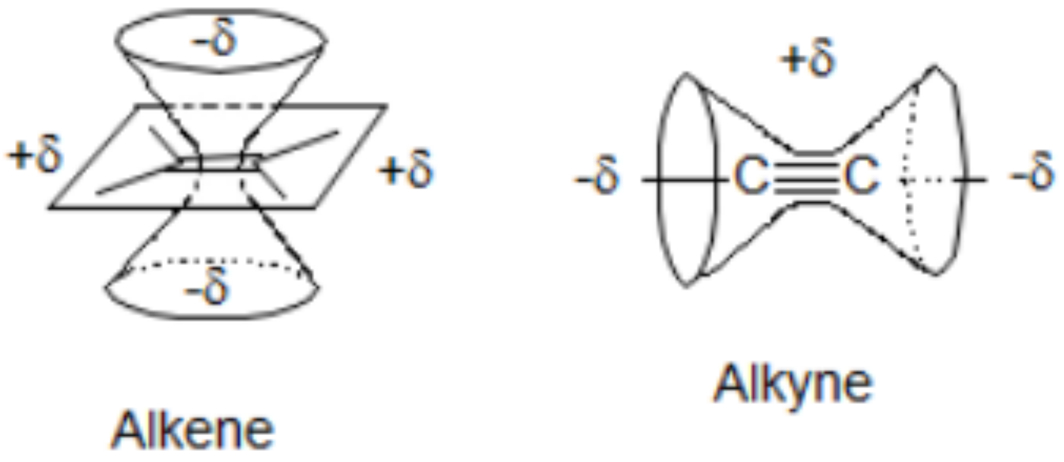
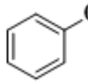
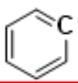
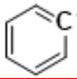


Figure adapted from (accessed June 8, 2020) https://www2.chem.wisc.edu/areas/reich/nmr/05-hmr-02-delta.htm

Characteristic Chemical Shifts

Characteristic NMR Chemical Shifts

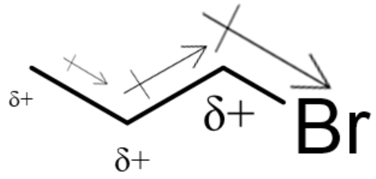
Structural Residue	Name	^1H δ (ppm)
R-CH_3	Primary alkyl	~ 0.9
R_2-CH_2	Secondary alkyl	~ 1.3
R_3-CH	Tertiary alkyl	~ 1.5
R_4C	Quaternary alkyl	N.A.
C=C-CH	Allylic	~ 1.7
R-C(=O)-CH	α to Carbonyl	2 – 2.7
C≡C-H	Alkynyl	2 – 3
CH-I	Alkyl Iodide	2 – 4
 CH	Benzylic	2.2 – 3
CH-N	Amine	2.4 – 3.4
CH-Br	Alkyl Bromide	2.5 – 4
CH-Cl	Alkyl Chloride	3 – 4
CH-O	Alcohol or Ether	3.3 – 4
CH-F	Alkyl Fluoride	4 – 4.5
R-OH	Hydroxyl	0.5 – 5 (often broad)
R-NH	Amino	0.5 – 5 (often broad)
 C-OH	Phenol	4 – 8
C=C-H	Alkenyl (Vinyl)	4.5 – 6
R-C(=O)-NH	Amide	5 – 8
 C-H	Aromatic	6.5 – 8
R-C(=O)-H	Aldehyde	9 – 10
R-C(=O)-OH	Carboxyl	10 – 12

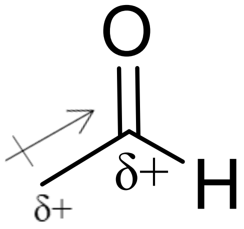
Estimating Relative Electron Density

Inductive withdrawing effect

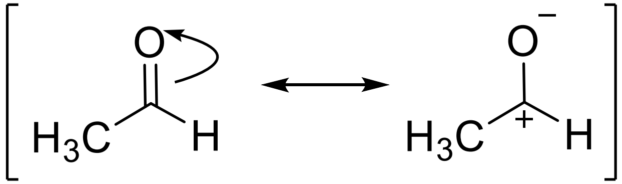
Withdraws electron density through σ (sigma) bonds

- Electronegative elements
- Nearby positive or partial positive charges





Positive Charge
"Downfield"
"Deshielded"
Higher chemical shift!



Estimating Relative Electron Density

Inductive withdrawing effect

Withdraws electron density through σ (sigma) bonds

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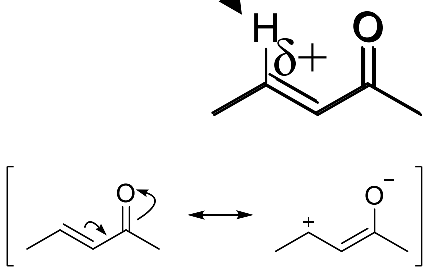
Resonance withdrawing effect

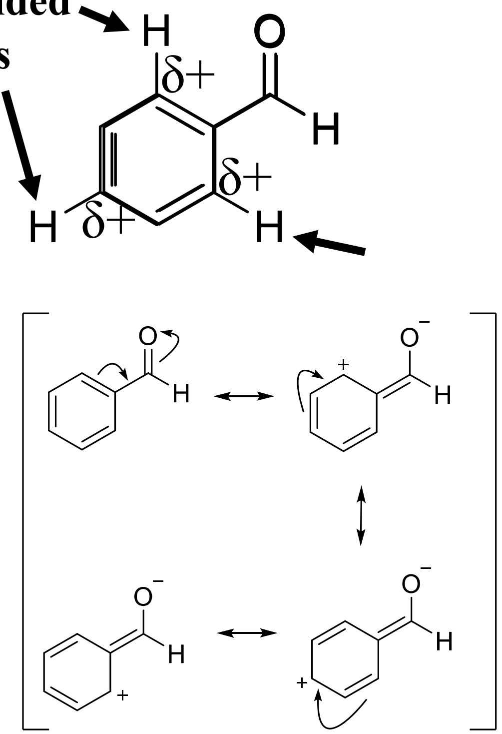
Conjugated π (pi) systems

- Resonance structures lead to + (positive) charges on atoms with attached hydrogens

Deshielded proton



Deshielded protons



Positive Charge

"Downfield"

"Deshielded"

Higher chemical shift!

Estimating Relative Electron Density

Inductive withdrawing effect

Withdraws electron density through σ (sigma) bonds

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- Nearby positive or partial positive charges

Resonance withdrawing effect

Conjugated π (pi) systems

- Resonance structures lead to + (positive) charges on atoms with attached hydrogens

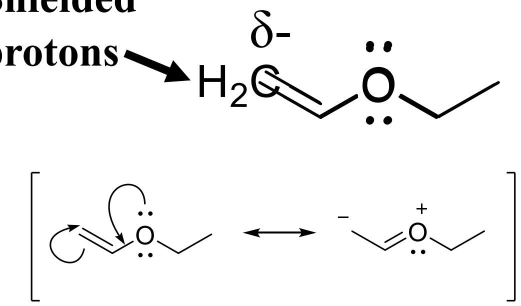
Resonance donating effect

Conjugated π (pi) systems

- Resonance structures lead to - (negative) charges on atoms with attached hydrogens
- Hint: look for lone pairs on heteroatoms (e.g. O, N) directly attached to π systems!

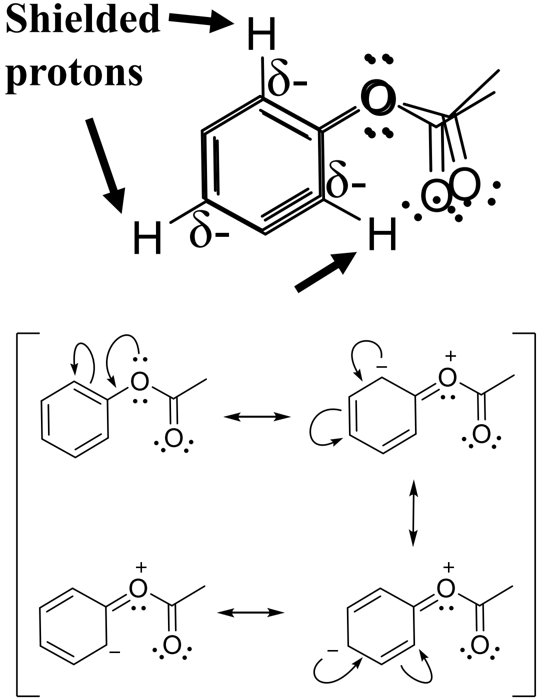
(Halogens are poor π donors, their inductive withdrawing effect is stronger)

Shielded protons



Negative Charge
“Upfield”
“Shielded”
Lower chemical shift!

Shielded protons



Video Summary

Chemical Shift depends on chemical environment

- **Deshielded protons (less electron density) appear downfield at high chemical shifts**
- **Shielded protons (more electron density) appear upfield at low chemical shifts**
- **Can draw resonance structures and use electronegativity arguments to predict shielding/deshielding effects**
- **Practice using the "Characteristic NMR Chemical Shifts Table"!**

Next video: Signal splitting patterns and coupling constants